FOSFOMYCIN

Single-dose cystitis

Pregnancy: compatible

CHEMICAL NAME [(2R,3S)-3-methyloxiran-2-yl]phosphonic acid (as trometamol or disodium salt)

CLASS Phosphonic acid derivative antibiotic

TRADE NAMES Monurol · Monuril · Fosfocin (IV)

THERAPEUTIC USES / INDICATIONS

- Acute uncomplicated cystitis in women (single dose).
- Cystitis caused by ESBL-producing *E. coli* (multidrug resistant).
- IV: severe multidrug-resistant infections (combination, specialist).
- UTI prophylaxis before urological procedures.

PHARMACODYNAMICS

- Inhibits **MurA** (UDP-N-acetylglucosamine enolpyruvyl transferase) → blocks first step of cell-wall synthesis → bactericidal.
- Unique mechanism → little cross-resistance.

ADMINISTRATION

- **PO** granules: dissolve 3 g sachet in half a glass of water and drink immediately.
- Take on an empty stomach, preferably at bedtime after emptying bladder.
- **IV** (disodium) — high sodium load.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea.
- Headache, vulvovaginitis.
- Dyspepsia.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis (rare).
- *C. difficile* colitis.
- IV: hypernatraemia, hypokalaemia, heart failure (sodium load).

DOSAGES

- **Adult cystitis:** 3 g single dose.
- **Complicated/recurrent (some guidelines):** 3 g every 72 h for 3 doses.
- **Procedure prophylaxis:** 3 g 3 h before.
- **IV:** 12–24 g/day in 2–3 doses (specialist).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Severe renal impairment (CrCl <10 mL/min) — oral form.
- Haemodialysis.

KEY INTERACTIONS

- **Metoclopramide:** ↓ fosfomycin levels.
- **Food:** delays absorption.

PHARMACOKINETICS

- **Peak:** 2 h. Bioavailability 35–40%.
- High urinary levels maintained for 24–48 h.
- Excreted unchanged in urine. **t½:** 4–8 h.

NURSING MANAGEMENT

- Explain how and when to take the sachet.
- Send urine culture before dosing if recurrent or complicated.
- Monitor symptoms; expect improvement in 2–3 days.
- IV: monitor sodium, potassium and fluid balance.

PATIENT EDUCATION

- Dissolve sachet in water and drink at bedtime after emptying your bladder.
- Symptoms may take 2–3 days to settle.
- Return if symptoms persist or fever/back pain develops.

KEY POINTS

- Single 3 g dose for cystitis.
- Unique MurA inhibitor.
- Active against ESBL E. coli.
- Empty stomach, bedtime.